



Spark Partitions

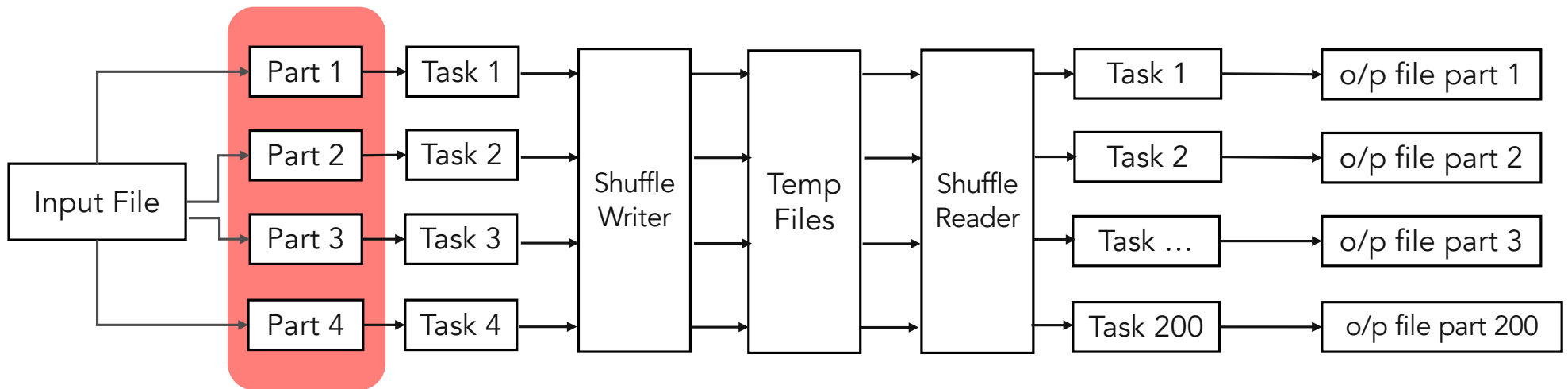
Input, Shuffle, Output

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Agenda

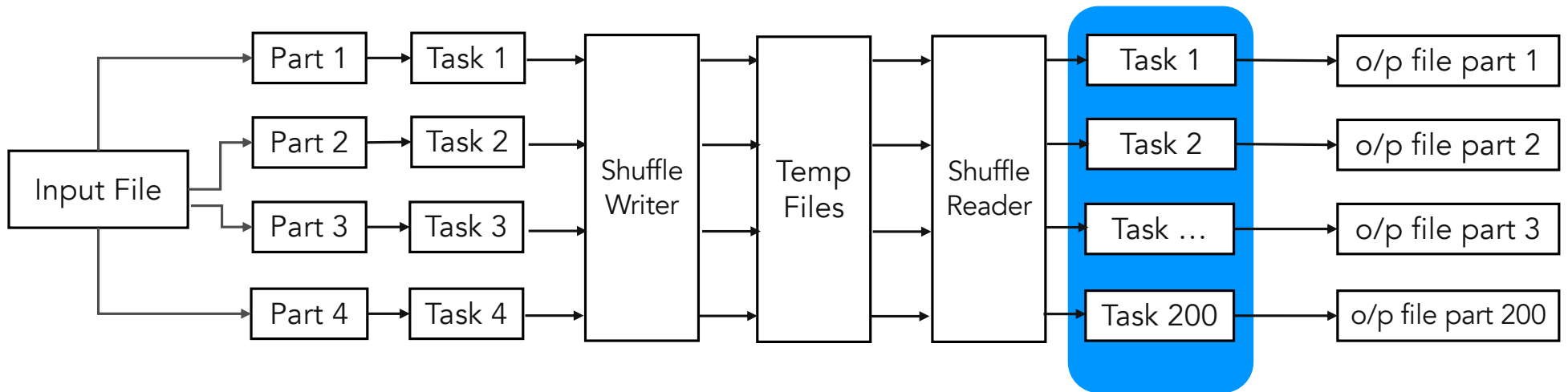
1. Input Partitions
2. Experiment using maxPartitionBytes
3. Shuffle Partitions
4. Output Partitions
5. Repartition()
6. Coalesce()
7. PartitionBy()
8. Transformations that change Partitions
9. Output Mode

Input Partition



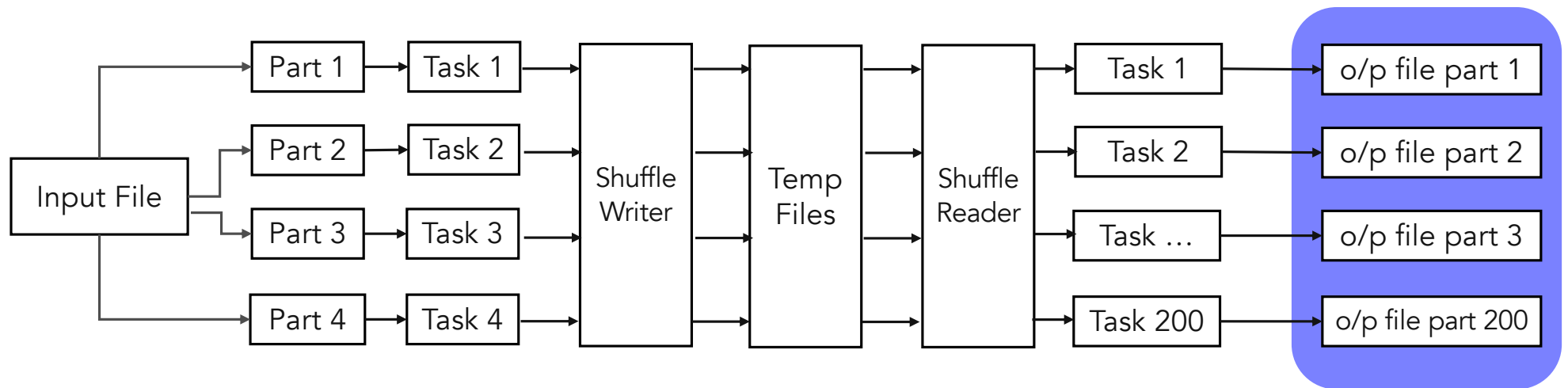
- Single partition size = HDFS block size = 128 MB
- No. of input partitions = File Size / 128 MB
- `spark.files.maxPartitionBytes`
- `spark.conf.set("spark.files.maxPartitionBytes", "536870912")`
- `df.rdd.getNumPartitions`

Shuffle Partition



- `spark.sql.shuffle.partitions`
- `spark.conf.set("spark.sql.shuffle.partitions", "20")`

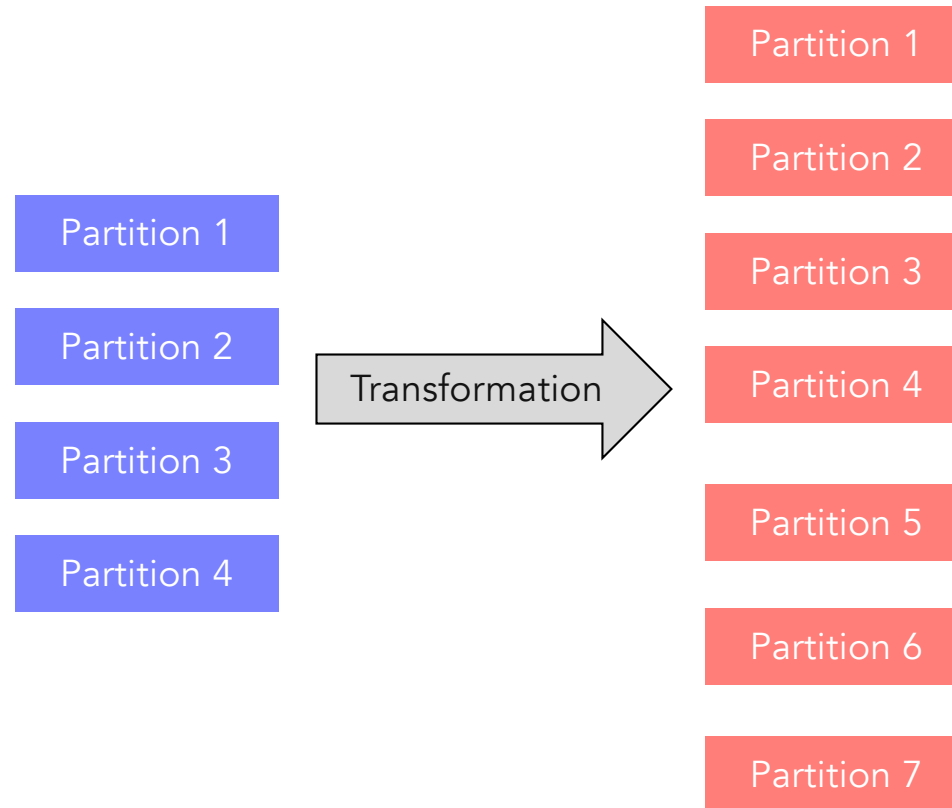
Output Partition



- `DataFrame.repartition(5)` – no. of partitions can be increased or decreased.
- `DataFrame.repartition("<column1>", "<column2>")`
- `DataFrame.coalesce(5)` – no. of partitions can only be decreased.
- `DataFrame.repartition(<no-of-partition>, "<column2>")`
- `DataFrame.write...partitionBy("<column1>", "<column2>")`

Partition Changing Transformations

- What is change in Partitions?
- Experiment
 - Select **No**
 - Union **Yes**
 - Subtract **Yes**
 - intersect **Yes**
 - sort **Yes**
 - selectExpr



File Write Mode

- DataFrame Writer Modes:
 - Overwrite
 - Append
 - Ignore
 - Error



Thank You
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